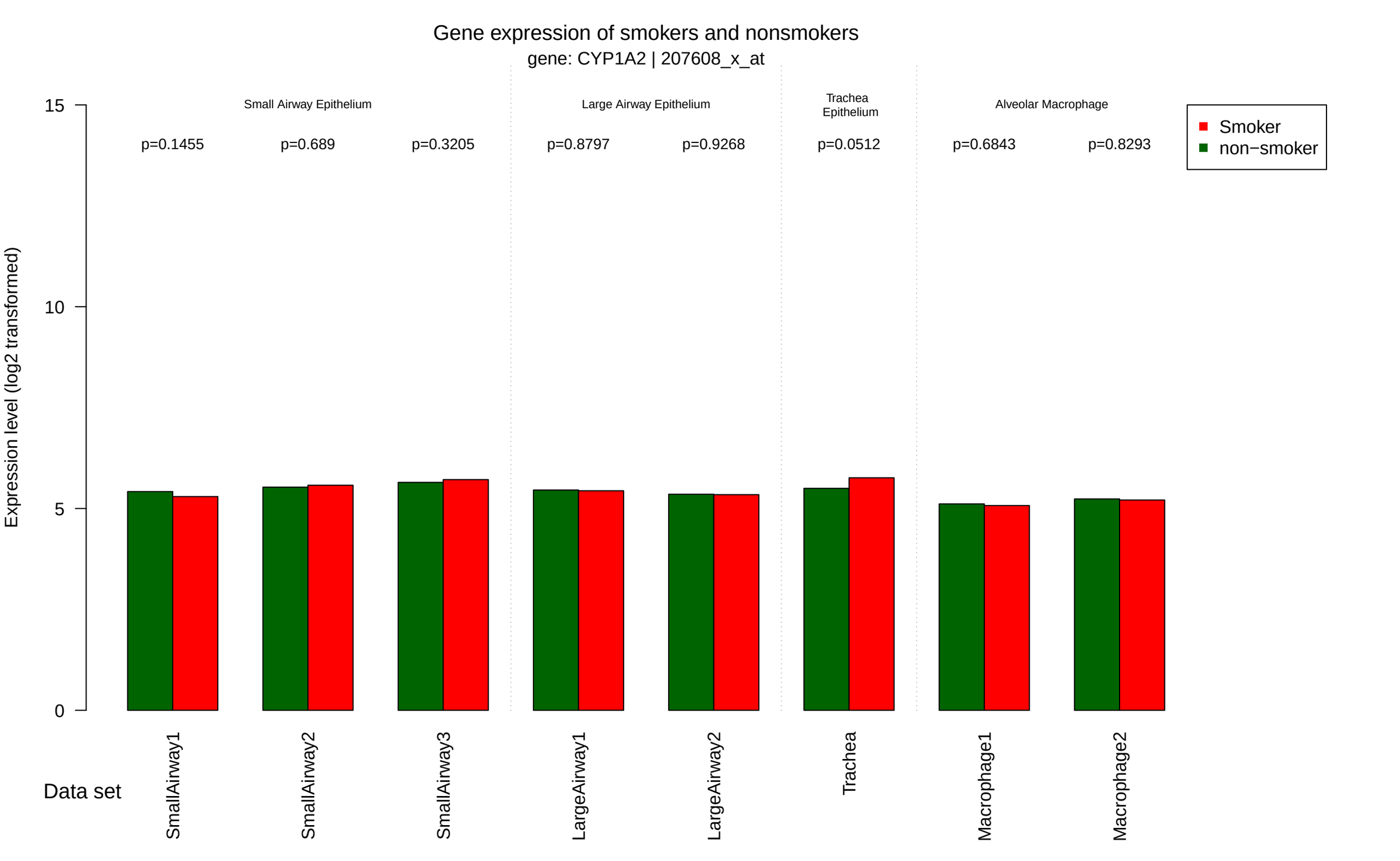
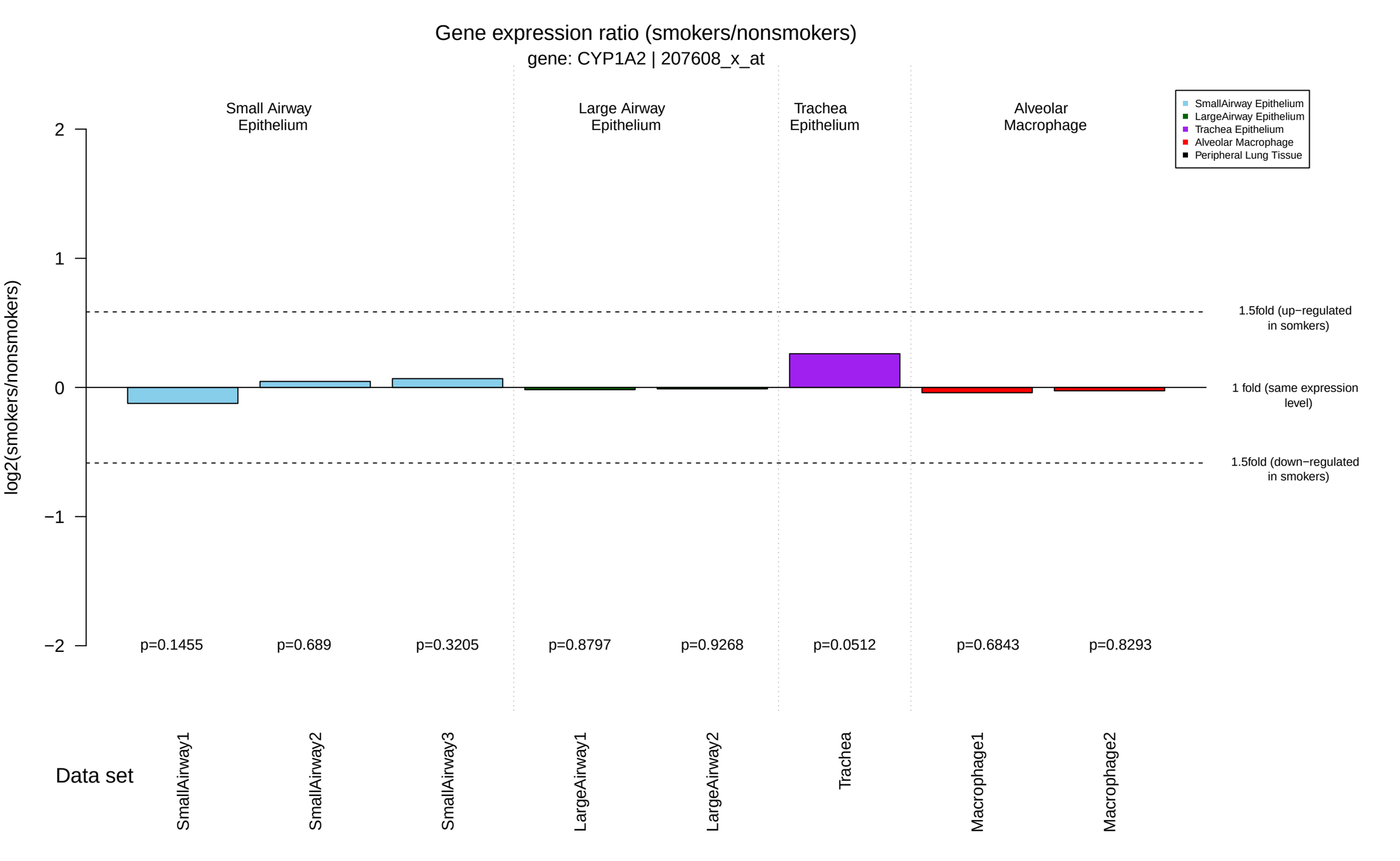
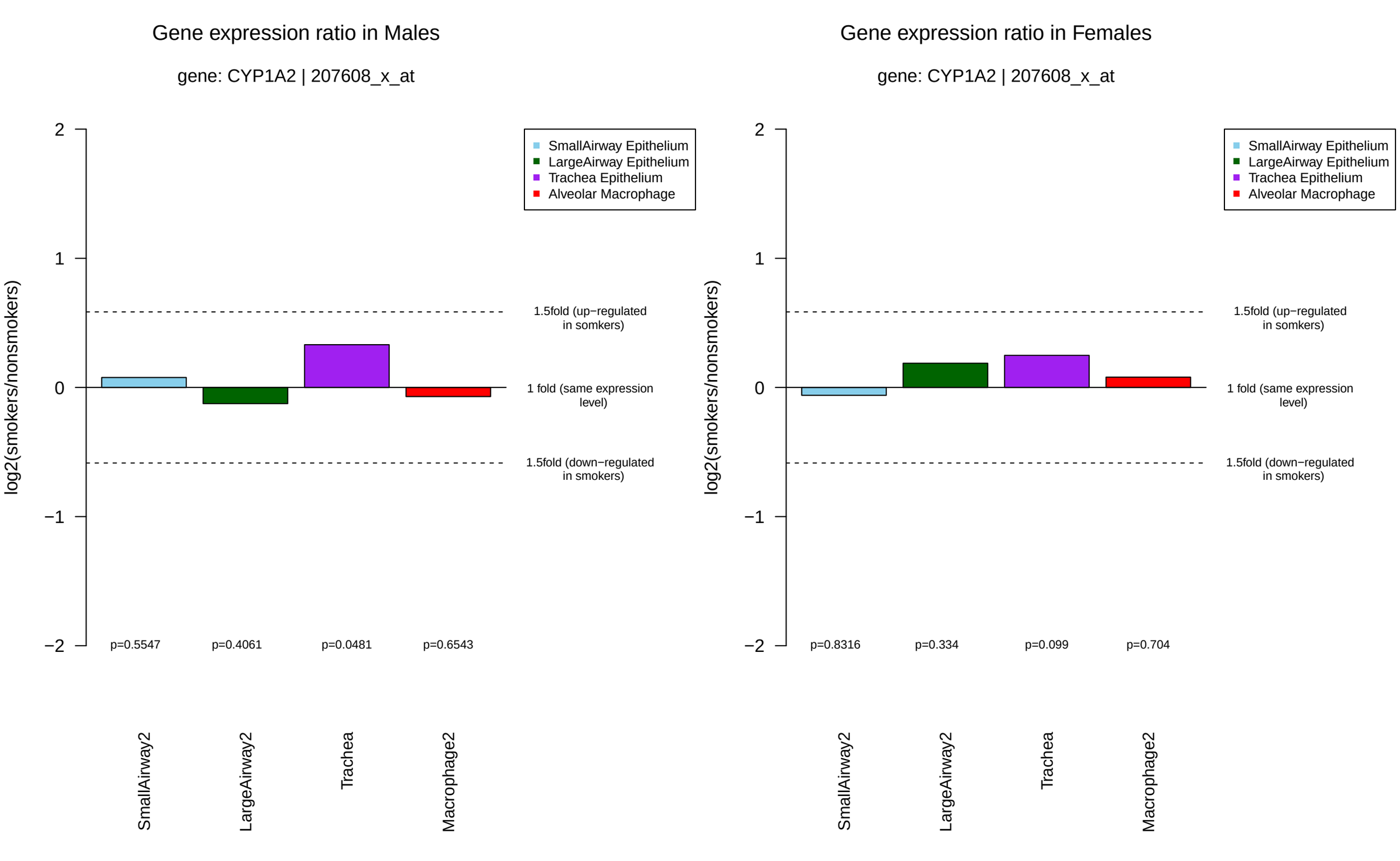
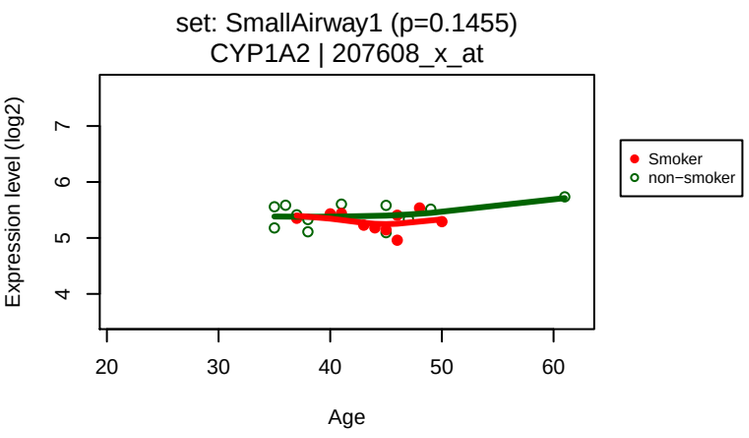
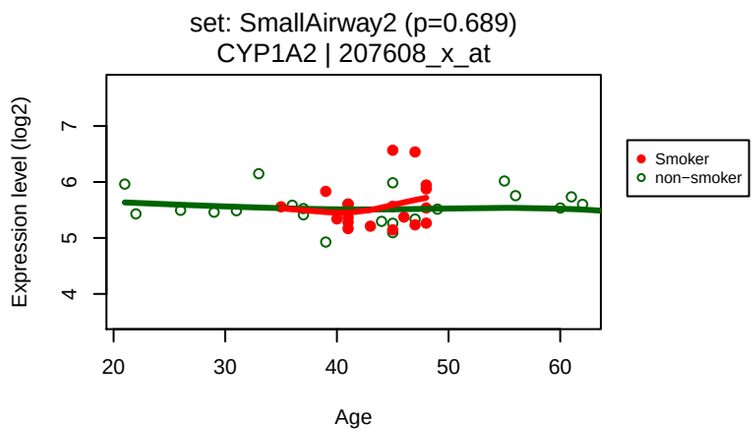
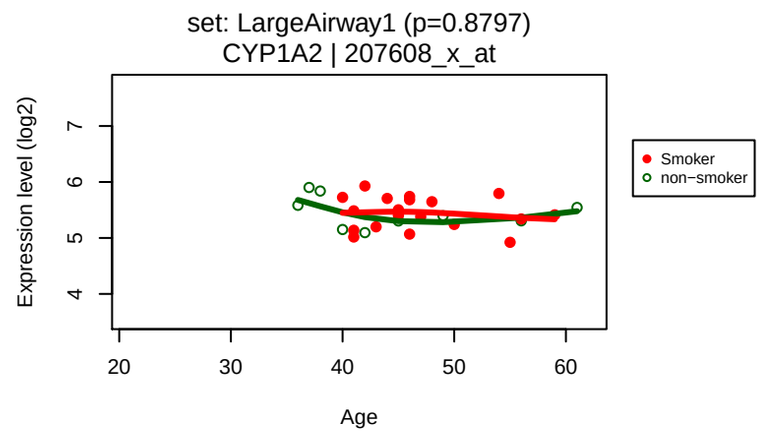

Gene expression of smokers and nonsmokers
gene: CYP1A2 | 207608_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1455
p=0.689
p=0.3205
p=0.8797
p=0.9268
p=0.0512
p=0.6843
p=0.8293
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

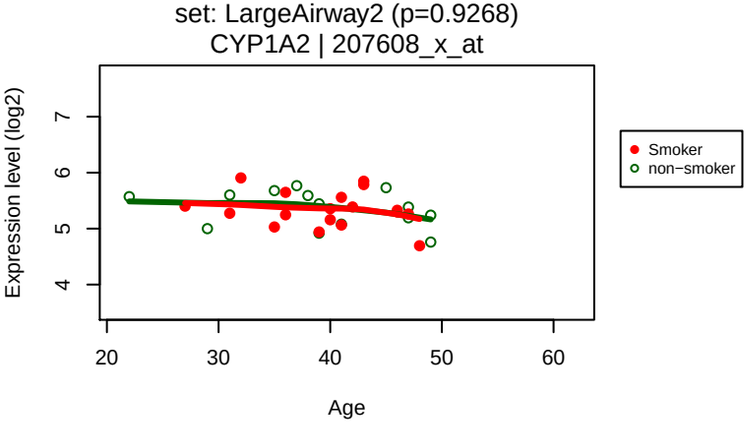

Gene expression ratio (smokers/nonsmokers)
gene: CYP1A2 | 207608_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1455
p=0.689
p=0.3205
p=0.8797
p=0.9268
p=0.0512
p=0.6843
p=0.8293
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

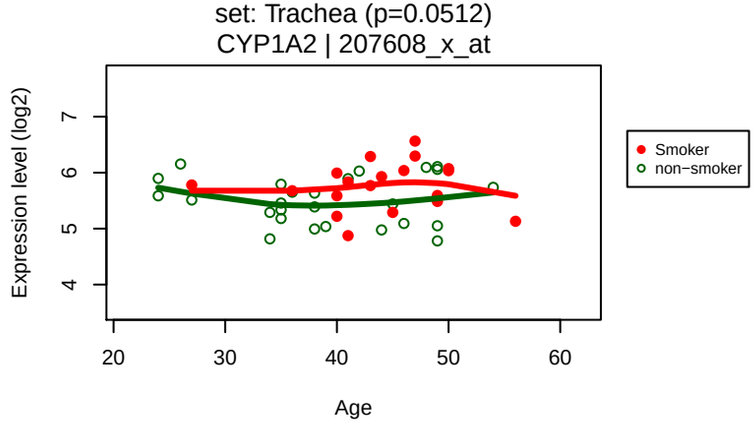

Gene expression ratio in Males
gene: CYP1A2 | 207608_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5547
p=0.4061
p=0.0481
p=0.6543
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP1A2 | 207608_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8316
p=0.334
p=0.099
p=0.704
SmallAirway2
LargeAirway2
Trachea
Macrophage2

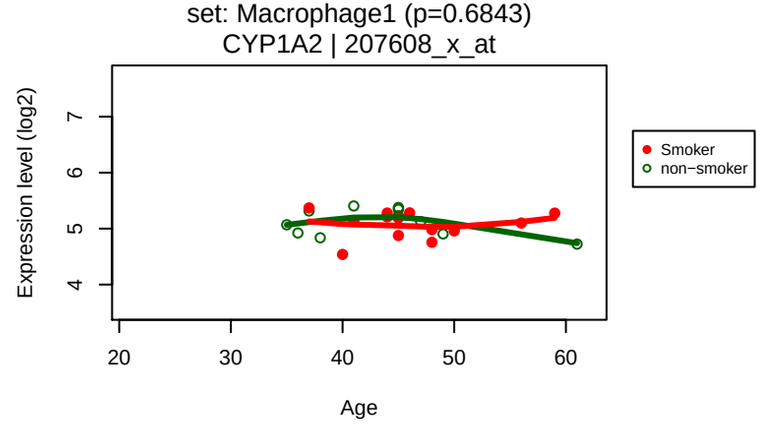

set: SmallAirway1 (p=0.1455)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker

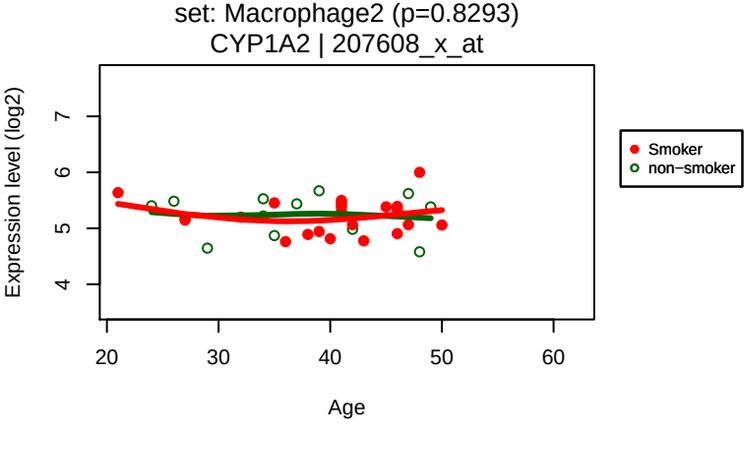

set: SmallAirway2 (p=0.689)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker

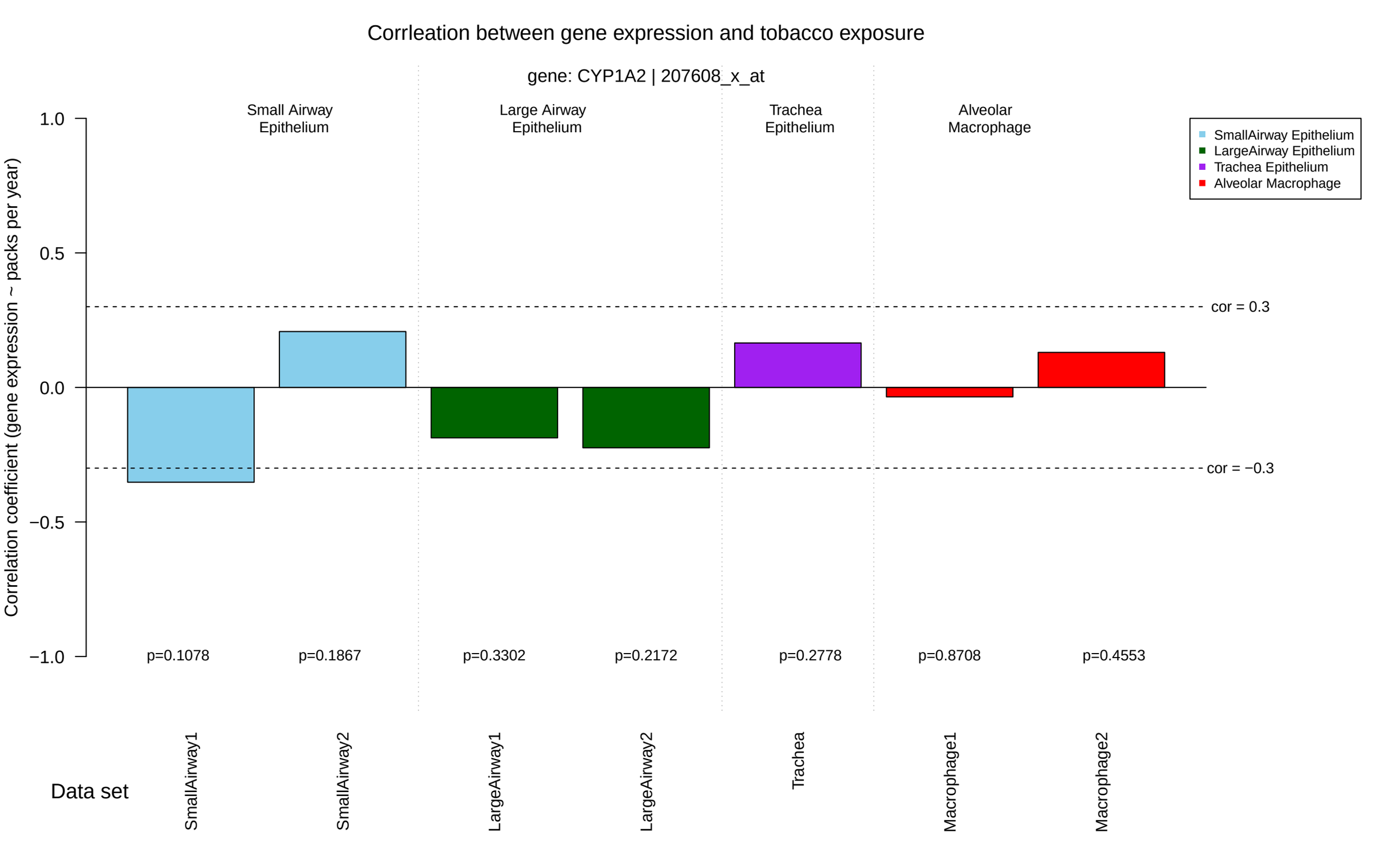

set: LargeAirway1 (p=0.8797)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9268)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0512)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6843)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8293)
CYP1A2 | 207608_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP1A2 | 207608_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1078
p=0.1867
p=0.3302
p=0.2172
p=0.2778
p=0.8708
p=0.4553
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2